



agriPairMetaFlow 0.1.0



Infographic 1 — Core Workflow Overview

Foundations for treatment-versus-control meta-analysis in agronomy



1 What 0.1.0 delivers



Foundational workflow for paired treatment-versus-control evidence synthesis.



Data import, validation, auditing, planning, effect-size calculation, model fitting, heterogeneity, prediction, and basic visualization.



Designed for agronomic experiments and clear interpretation.

2 Core functions in this release

`f()` `apm_read()`

`f()` `apm_validate()`

`f()` `apm_audit()`

`f()` `apm_plan()`

`f()` `apm_recover_uncertainty()`

`f()` `apm_effect_size()`

`f()` `apm_fit()`

`f()` `apm_subgroup()`

`f()` `apm_heterogeneity()`

`f()` `apm_prediction()`

`f()` `apm_threshold()`

`f()` `apm_forest()`

`f()` `apm_funnel()`

`f()` `apm_table()`

3 Typical workflow



4 Applied agronomy example



Studies compare a treatment against a control.



Example outcome: yield response.



Effect size: log response ratio (lnRR).



Goal: estimate average effect and uncertainty across studies.

6 Mini code sketch

```
dat <- apm_read('meta.xlsx')
val <- apm_validate(dat)
audit <- apm_audit(dat)
plan <- apm_plan(dat)
es <- apm_effect_size(plan, measure = 'lnRR')
fit <- apm_fit(es)
apm_forest(fit)
```

5 Interpretation essentials



Distinguish mean effect from heterogeneity. The mean effect summarizes the average response; heterogeneity quantifies between-study variation.



Prediction intervals help assess expected performance in new studies. They reflect both within-study uncertainty and between-study heterogeneity.



Threshold analysis helps interpret practical relevance. Compare effects to agronomic relevance thresholds to inform decisions.

7 Take-home message



- ✓ Version 0.1.0 establishes the **full basic pipeline**.
- ✓ It is the **entry point** before dependence-aware and robust extensions.





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Infographic 2 — Data Ingestion, Validation, and Audit

Build a trustworthy dataset before calculating effect sizes



1 Main goal



Import data consistently.



Validate structure and required fields.



Audit potential problems before modeling.



Define the estimand and study structure.

2 Functions emphasized

`f()` `apm_read()`

`f()` `apm_validate()`

`f()` `apm_audit()`

`f()` `apm_plan()`

3 What to validate

- ✓ study identifiers
- ✓ treatment and control labels
- ✓ means, SD/SE, and sample sizes
- ✓ missing values
- ✓ duplicated contrasts
- ✓ compatible units
- ✓ positive means for ratio-based measures

4 What the audit should reveal



impossible SD values



unclear control groups



reused controls



inconsistent coding across studies



mixed scales or units



missing uncertainty information

5 Planning the estimand



`apm_plan()` organizes the analysis around:

- study
- experiment
- treatment
- control
- response variable
- optional moderators

6 Applied agronomy example



Maize fertilizer trials from multiple locations and seasons.

Response variable: yield (t/ha). Compare each fertilizer treatment (e.g., N, NP, NPK, organic) against the study-specific control.

Several treatment-versus-control comparisons across studies.

7 Mini code sketch

```
dat <- apm_read('fertilizer_trials.xlsx')
val <- apm_validate(dat)
audit <- apm_audit(dat)
plan <- apm_plan(
  dat,
  study = study_id,
  treatment = treatment,
  control = control,
  response = yield
)
```

8 Take-home message



- ✓ Good meta-analysis starts with clean structure.
- ✓ Validation and audit reduce downstream errors.
- ✓ A clear plan makes later modeling defensible.





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Infographic 3 — Uncertainty Recovery and Effect Sizes

Translate treatment-versus-control data into analyzable contrasts



1 Core purpose



Recover uncertainty when studies report SD, SE, CI, CV, or related summaries.



Compute effect sizes suitable for agronomic meta-analysis.

2 Functions emphasized

f()

`apm_recover_uncertainty()`

f()

`apm_effect_size()`

3 Common effect sizes



lnRR: log response ratio



MD: mean difference



SMD: standardized mean difference



VR: variability ratio



CVR: coefficient-of-variation ratio



RR / OR / RD for binary outcomes

4 Mini formulas

lnRR

$$\ln RR = \ln \left(\frac{\text{mean}_T}{\text{mean}_C} \right)$$

MD

$$MD = \text{mean}_T - \text{mean}_C$$

VR

$$VR = \ln \left(\frac{SD_T}{SD_C} \right)$$

5 How uncertainty can be recovered



SE → SD



CI → SE or SD



CV + mean → SD



n + SD or SE →
variance of the effect size

6 Applied agronomy examples



Yield response to fertilizer: lnRR



Disease incidence under seed treatment: RR



Stability under irrigation management: VR or CVR

7 Mini code sketch

```
unc <- apm_recover_uncertainty(dat)
es <- apm_effect_size(
  unc,
  measure = 'lnRR',
  treatment_mean = mean_t,
  control_mean = mean_c,
  treatment_sd = sd_t,
  control_sd = sd_c,
  treatment_n = n_t,
  control_n = n_c
)
```

8 Take-home message



Choose the effect size that matches the scientific question.



Ratios are often intuitive in agronomy.



Recovered uncertainty must remain transparent.





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Infographic 4 — Model Fitting, Heterogeneity, and Prediction

Summarize average response while keeping variation visible



1 Functions emphasized



`apm_fit()`



`apm_subgroup()`



`apm_heterogeneity()`



`apm_prediction()`



`apm_threshold()`

2 What the model returns



pooled effect



standard error



confidence interval



heterogeneity estimates



prediction interval

3 Heterogeneity essentials

Q

Cochran's Q

τ^2

tau-squared
between-study
variance

τ

tau
between-study
SD

I^2

I-squared
proportion of
variation

H^2

H-squared
variability
ratio

These statistics quantify between-study variation across effect sizes.

4 Subgroups and context



- Compare effects across crop type, soil class, or management group.
- Use subgroup analysis to explore patterns, not to replace scientific judgment.

5 Prediction and thresholds



Prediction interval: likely range for a new study.



Threshold analysis: compare effects with agronomic relevance targets.

6 Applied agronomy example



Study: Irrigation meta-analysis across maize, wheat, and rice.



Outcome: yield response.



Question: average benefit and expected performance in a new environment.



Maize



Wheat



Rice

7 Mini code sketch

```
fit <- apm_fit(es)
het <- apm_heterogeneity(fit)
pred <- apm_prediction(fit)
thr <- apm_threshold(fit, threshold = 0.05)
sub <- apm_subgroup(fit, by = crop)
```

8 Take-home message



- ✓ Mean effect is not the whole story.
- ✓ Heterogeneity and prediction intervals are essential.
- ✓ Thresholds improve practical interpretation.





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Infographic 5 — Forest Plots, Funnel Plots, and Tables

Turn the fitted model into clear scientific outputs



1 Functions emphasized



`apm_forest()`



`apm_funnel()`



`apm_table()`

2 Forest plot: what it shows



study-level effects



confidence intervals



pooled estimate



visual weight and consistency

3 Funnel plot: what it shows



effect size versus precision



possible small-study patterns



visual starting point, not proof of bias

4 Tables: what they summarize



model estimates



heterogeneity metrics



subgroup summaries



prediction information



export-ready outputs

5 Good reporting habits



report the effect scale clearly



include confidence and prediction intervals



label outcomes and moderators explicitly



keep tables aligned with the plotted result

6 Applied agronomy example

Biochar versus control across soil-carbon studies:



forest plot
for pooled effect

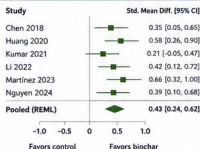


funnel plot
for small-study inspection



table
for manuscript-ready summary

Forest plot (pooled effect)



Funnel plot (small-study inspection)

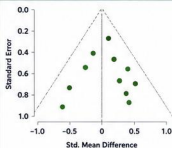


Table (manuscript-ready summary)

Outcome: Soil organic carbon	Estimate (SMD)	95% CI	Prediction Interval
Pooled effect (REML)	0.43	[0.24, 0.62]	[-0.02, 0.88]
Tau ²	0.07	—	—
I ²	58%	—	—
Q (df)	12.4 (5)	—	—
Studies (k)	6	—	—

Effect scale: Standardized Mean Difference (Hedges' g)

7 Mini code sketch



```
apm_forest(fit)
```

```
apm_funnel(fit)
```

```
out_tab <- apm_table(fit)
```

8 Take-home message



- ✓ Plots support interpretation.
- ✓ Tables support reproducible reporting.
- ✓ Clear outputs make methods easier to trust and teach.





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Infographic 6 — Dependence-Aware Meta-Analysis Overview

Why version 0.2.0 matters when effects are not independent



1 What 0.2.0 adds



dependence-aware workflow



shared controls



paired covariance structures



multilevel models



robust inference



wild bootstrap sensitivity

2 Functions in this release

`f()` `apm_shared_control()`

`f()` `apm_vcov()`

`f()` `apm_pair_vcov()`

`f()` `apm_dependence_audit()`

`f()` `apm_rho_sensitivity()`

`f()` `apm_multilevel()`

`f()` `apm_variance_components()`

`f()` `apm_robust()`

`f()` `apm_wild_bootstrap()`

`f()` `apm_compare_inference()`

3 Why dependence occurs



multiple treatments
share one control



repeated outcomes
from one study



paired or correlated
designs



multiple effect sizes
from the same paper

4 Typical 0.2.0 workflow



1
Detect
dependence



2
Build V



3
Audit
dependence



4
Fit multilevel
model



5
Quantify
variance
components



6
Apply robust
inference



7
Compare
conclusions

5 Applied agronomy example



Nitrogen-rate trials
with 50, 100, and
150 kg N ha⁻¹ all
compared with the
same zero-N control.

6 Take-home message



✓ Ignoring dependence
can overstate precision.

✓ Version 0.2.0 explicitly
models correlated
effect sizes.





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Infographic 7 — Shared Controls and Covariance Matrices

Build the V matrix instead of pretending all contrasts are independent



1 Functions emphasized



`apm_shared_control()`

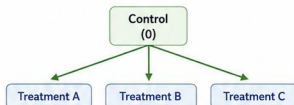


`apm_vcov()`



`apm_pair_vcov()`

2 The shared-control problem



One control group is used in multiple contrasts. Each contrast reuses information from the same control, so the effect sizes are not independent.

3 What the covariance matrix does



Places study-specific variances on the diagonal.



Adds covariances off the diagonal.



Preserves the dependence structure among effect sizes.

4 Paired designs



Paired treatment-control observations may require correlation-aware variance formulas.



The paired structure differs from shared-control multigroup dependence.

5 Applied agronomy example



6 Mini code sketch

```

sc <- apm_shared_control(es)

V <- apm_vcov(es)

Vp <- apm_pair_vcov(es, rho = 0.5)
  
```

7 Take-home message



- ✓ A correct **V** matrix is central to dependence-aware synthesis.
- ✓ Shared controls and paired designs are related but not identical problems.





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Infographic 8 — Dependence Audit and Rho Sensitivity

Check what is known, and test what is uncertain



1 Functions emphasized

f() `apm_dependence_audit()`

f() `apm_rho_sensitivity()`

3 Why rho matters

ρ

Some paired or correlated effects require an assumed correlation rho.

If rho is uncertain, conclusions should be checked across plausible values.

2 What the dependence audit checks



repeated study IDs



reused controls



possible paired structures



duplicated contrasts



missing correlation information



unclear clustering levels

4 Sensitivity concept



5 Applied agronomy example



Repeated measurements of plant response or paired block comparisons where the treatment-control correlation is not reported.



We audit for dependence issues and test how results change across assumed rho values.

7 Take-home message



- ✓ Audit first, then model.
- ✓ Sensitivity analysis is better than silently fixing rho.
- ✓ Stable conclusions increase credibility.

6 Mini code sketch

```
da <- apm_dependence_audit(es)

rho_scan <- apm_rho_sensitivity(
  es,
  rho = seq(0, 0.9, by = 0.1)
)
```





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Infographic 9 — Multilevel Models and Variance Components

Partition heterogeneity when studies contain multiple effects



1 Functions emphasized

`f()` `apm_multilevel()`

`f()` `apm_variance_components()`

2 Why multilevel modeling?



one paper may contain several experiments



one experiment may produce several effects



effect sizes within a study are often correlated

3 Typical hierarchy



Study



Experiment



Effect size



Optional note: site and year can also create levels.

4 What variance components tell you



between-study heterogeneity



within-study or within-experiment heterogeneity



how variation is distributed across levels

5 Applied agronomy example



A paper reports multiple crops, sites, or years, each contributing several treatment-versus-control contrasts.

6 Mini code sketch

```
m1 <- apm_multilevel(
  es,
  random = ~ 1 | study_id /
    experiment_id / effect_id
)
vc <- apm_variance_components(m1)
```

7 Take-home message



Multilevel models respect hierarchical evidence structures.



Variance components help explain where heterogeneity comes from.





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Infographic 10 — Robust Inference and Wild Bootstrap

Stress-test conclusions when dependence and few clusters make inference harder



1 Functions emphasized

`f()` `apm_robust()`

 `apm_wild_bootstrap()`

 `apm_compare_inference()`

2 Why robust inference?



conventional SEs may be optimistic



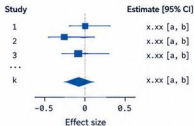
few studies or clustered effects can distort uncertainty



CR2 and wild bootstrap provide sensitivity checks

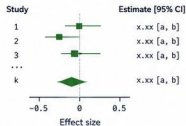
3 Three views of inference

Conventional model-based inference



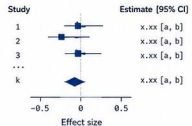
 Assumes independent effects and large-sample SEs.

Cluster-robust CR2 inference



 Adjusts SEs for clustering using CR2 small-sample correction.

Wild bootstrap inference



 Uses wild bootstrap to approximate the sampling distribution.

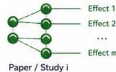


Compare conclusions across methods. Disagreement signals sensitivity to dependence or small-sample issues.

4 What to compare

-  pooled estimate
-  standard error
-  confidence interval
-  p-value or significance conclusion
-  practical stability of the result

5 Applied agronomy example



A meta-analysis with several correlated effects per paper and a limited number of independent studies.

6 Mini code sketch

```
rob <- apm_robust(m1, cluster = study_id)
wb <- apm_wild_bootstrap(m1, cluster = study_id)
cmp <- apm_compare_inference(
  conventional = m1,
  robust = rob,
  wild = wb
)
```

7 Take-home message



- ✓ Robust inference does **not** replace the main model.
- ✓ It shows whether conclusions are **sensitive** to dependence and small-sample issues.





agriPairMetaFlow 0.3.0



Infographic 11 — Moderator Analysis and Dose-Response Overview

Explain why effects differ across agronomic conditions and input levels



1 What 0.3.0 adds

-  moderator screening
-  meta-regression
-  interactions
-  marginal effects
-  curve-based meta-regression
-  dose-response modeling
-  curve feature extraction
-  comparison of competing moderator models

2 Functions in this release

- | | |
|--|--|
| <code>f()</code> <code>apm_moderator_screen()</code> | <code>f()</code> <code>apm_dose_response()</code> |
| <code>f()</code> <code>apm_metareg()</code> | <code>f()</code> <code>apm_curve_features()</code> |
| <code>f()</code> <code>apm_interaction()</code> | <code>f()</code> <code>apm_bubble()</code> |
| <code>f()</code> <code>apm_marginal_effects()</code> | <code>f()</code> <code>apm_dose_plot()</code> |
| <code>f()</code> <code>apm_metareg_curve()</code> | <code>f()</code> <code>apm_model_compare()</code> |

3 Why moderators matter



Studies differ in climate, soil, cultivar, dose, and management.



Meta-regression relates effect sizes to these covariates.



Interpretation should focus on patterns, not just p-values.

4 Typical 0.3.0 workflow



1
Screen moderators



2
Fit meta-regression



3
Compare models



4
Examine interactions



5
Estimate marginal effects



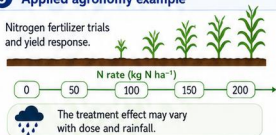
6
Fit dose-response curve



7
Visualize and interpret

5 Applied agronomy example

Nitrogen fertilizer trials and yield response.



6 Take-home message



- ✓ Version 0.3.0 explains heterogeneity with moderators.
- ✓ Dose-response curves support practical recommendations.
- ✓ Visual summaries improve interpretation and decisions.

7 Mini code sketch

```
mods <- apm_moderator_screen(es)
mr <- apm_metareg(es, mods = ~ rainfall + soil_ph)
curve <- apm_dose_response(es, moderator = dose_n)
apm_dose_plot(curve)
```





agriPairMetaFlow 0.3.0

Infographic 12 — Moderator Screening and Meta-Regression

Relate effect sizes to agronomic covariates in a structured way



1 Functions emphasized

`f()` `apm_moderator_screen()`

`f()` `apm_metareg()`

`f()` `apm_model_compare()`

2 What moderator screening does



checks candidate moderators



summarizes missingness and spread



flags low-information predictors



ranks promising covariates

3 Types of moderators



Categorical
e.g. crop type,
soil class



Continuous
e.g. rainfall,
pH



Design variables
e.g. year,
duration,
location

4 Meta-regression essentials



Effect size is
the response.



Moderators
explain
heterogeneity.



Coefficients
describe expected
change in effect.



Compare
models
cautiously.

5 Applied agronomy example

Irrigation response across
environments.

Candidate moderators:
rainfall and clay content.



Environment	Rainfall (mm)	Clay (%)	Effect Size
Site A	320	18	0.35
Site B	640	32	0.05
Site C	210	42	-0.25
...	...	...	...

6 Mini code sketch

```
scr <- apm_moderator_screen(es)
mr1 <- apm_metareg(es, mods = ~ rainfall)
mr2 <- apm_metareg(es, mods = ~ rainfall + clay)
apm_model_compare(mr1, mr2)
```

7 Workflow: 5 steps



1 Screen



2 Fit



3 Compare



4 Interpret



5 Report

7 Take-home message



- ✓ Moderator screening improves workflow discipline and data understanding.
- ✓ Meta-regression relates effect sizes to covariates and explains heterogeneity.
- ✓ Model comparison helps choose parsimonious models and avoid overfitting.





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Infographic 13 — Interactions, Marginal Effects, and Bubble Plots

Move beyond average slopes and visualize how moderators work together



1 Functions emphasized

f()	<code>apm_interaction()</code>	
f()	<code>apm_marginal_effects()</code>	
f()	<code>apm_bubble()</code>	

2 Why interactions matter



Fertilizer effect may depend on rainfall.



Cultivar response may depend on temperature.



Management responses can change across soil classes.

3 Marginal effects



Predicted effects (e.g., treatment effect) can be computed at chosen moderator values.



These predicted effects are easier to communicate and interpret than raw coefficients.

4 Bubble plot: what it shows



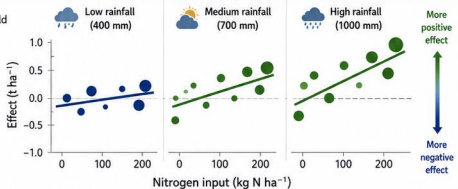
- One point per study or contrast.
- Bubble size reflects weight or precision.
- Fitted line shows overall trend.
- Color or symbol can indicate groups.

5 Applied agronomy example

Predicted effect of bioinoculant on yield (t ha⁻¹)

Bioinoculant trials show that the yield effect depends on both rainfall and nitrogen input.

- Point = study
- Size = weight
- Line = overall trend



6 Mini code sketch

```
int_fit <- apm_interaction(es,
  formula = ~ rainfall * dose_n)
me <- apm_marginal_effects(int_fit,
  at = c(400, 700, 1000))
apm_bubble(int_fit)
```

7 Take-home message



- ✓ Interactions reveal context dependence.
- ✓ Marginal effects simplify interpretation.
- ✓ Bubble plots connect models back to data.





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Infographic 14 — Dose-Response Meta-Analysis and Curve Features

Estimate how agronomic responses change across input levels



1 Functions emphasized

`f()` `apm_dose_response()`

`f()` `apm_metareg_curve()`

`f()` `apm_curve_features()`

`f()` `apm_dose_plot()`

2 What dose-response modeling answers

- ? is the response increasing or decreasing
- ? is there curvature
- ? where is the response plateau
- ? what range is practically useful

3 Common curve ideas

linear trend



saturation curve



hump-shaped response



threshold-like behavior



4 Curve features to extract

- slope direction
- peak or optimum
- plateau region
- response at user-defined doses

Workflow overview



Fit curve



Extract features



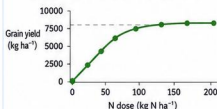
Plot



Interpret

5 Applied agronomy example

Nitrogen fertilizer dose (N) versus grain yield.



Yield increases rapidly at low N, then shows diminishing returns at higher doses, approaching a plateau.

6 Mini code sketch

```
dr <- apm_dose_response(
  es, moderator = dose_n)

cf <- apm_curve_features(
  dr, dose = c(0, 50,
    100, 150))

apm_dose_plot(dr)
```



Use the extracted features (cf) for reporting, decision support, and recommendations.

7 Take-home message

- curves are more informative than single-dose contrasts.
- extracted features support recommendations.
- plots make nonlinear patterns easier to see.





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Infographic 15 — Choosing and Comparing Moderator Models

Compare plausible explanations before drawing agronomic conclusions



1 Functions emphasized

`f()` `apm_metareg()`

`f()` `apm_dose_response()`

`f()` `apm_model_compare()`

`f()` `apm_dose_plot()`

2 Candidate models

 null model

 one-moderator model

 interaction model

 curve-based model

3 What to compare

 fit quality

 parsimony

 interpretability

 stability of conclusions

4 Model comparison workflow



1
Define
candidates



2
Fit
models



3
Compare
metrics



4
Inspect
plots



5
Select practical
interpretation

5 Applied agronomy example

Fertilizer-response alternatives:

Linear model



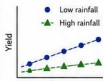
Constant change in
yield per unit N.

Curved model



Diminishing gains at
higher N rates.

Interaction with rainfall



Response depends on
rainfall conditions.

7 Comparison at a glance

Model	AICc (lower is better)	ΔAICc	Interpretability	Preferred
--- Null model	352.1	24.3	High	—
 One-moderator (rainfall)	331.3	3.5	High	—
 Curve-based (dose_n)	327.8	0.0	Medium	★ Preferred

ΔAICc = Difference from the best (lowest) AICc.

6 Mini code sketch

```
m0 <- apm_metareg(es, mods = ~ 1)
m1 <- apm_metareg(es, mods = ~ rainfall)
m2 <- apm_dose_response(es, moderator = dose_n)
apm_model_compare(m0, m1, m2)
```

7 Take-home message



- ✓ The best model is not always the most complex.
- ✓ Comparison supports transparent decisions.
- ✓ Practical relevance must guide final interpretation.





agriPairMetaFlow 0.4.0



Infographic 16 — Multivariate and Bayesian Meta-Analysis Overview

Model correlated outcomes and quantify uncertainty with richer probabilistic tools



1 What 0.4.0 adds

- multivariate models for correlated outcomes
- multivariate correlation sensitivity
- prior specification
- prior checking
- Bayesian model fitting
- Bayesian diagnostics
- Bayesian prediction
- Bayesian threshold analysis
- Bayesian model comparison
- multivariate visualization

2 Functions in this release

- | | |
|---|---|
| f() <code>apm_multivariate()</code> | f() <code>apm_bayes_predict()</code> |
| f() <code>apm_mvcor_sensitivity()</code> | f() <code>apm_bayes_threshold()</code> |
| f() <code>apm_prior()</code> | f() <code>apm_bayes_diagnostics()</code> |
| f() <code>apm_prior_check()</code> | f() <code>apm_bayes_compare()</code> |
| f() <code>apm_bayes()</code> | f() <code>apm_multivariate_plot()</code> |

3 Why 0.4.0 matters



Some agronomic studies report several related outcomes.



Outcome correlations should not be ignored.



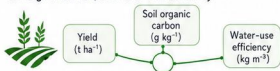
Bayesian analysis expresses uncertainty with full posterior distributions.

4 Typical 0.4.0 workflow



5 Applied agronomy example

Soil management studies jointly analyze yield, soil organic carbon, and water-use efficiency.



Model correlations and quantify uncertainty across all outcomes.

6 Take-home message



- ✓ Version 0.4.0 supports correlated outcomes.
- ✓ Bayesian workflows add transparent uncertainty assessment.
- ✓ Multivariate summaries improve agronomic interpretation.

7 Mini code sketch

```
mv <- apm_multivariate(es, outcomes = c("yield", "soc", "wue"))
pri <- apm_prior(mv)
fit_b <- apm_bayes(mv, prior = pri)
apm_multivariate_plot(fit_b)
```





agriPairMetaFlow 0.4.0

Infographic 17 — Multivariate Models and Correlated Outcomes

Analyze several related outcomes without pretending they are independent



1 Functions emphasized

f() `apm_multivariate()`

f() `apm_mvcor_sensitivity()`

f() `apm_multivariate_plot()`

2 Why multivariate models?



- one study may report several outcomes



- outcomes can be correlated



- joint modeling improves coherence

3 Typical outcome sets



Yield and soil organic carbon



Yield and nutrient uptake



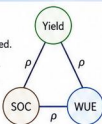
Biomass and water-use efficiency

4 Correlation sensitivity



Plausible cross-outcome correlations (ρ) can be checked.

Conclusions should be tested across different assumptions about ρ .



5 Applied agronomy example

Soil-management studies analyze three related outcomes:



Soil organic carbon (SOC)

Water-use efficiency (WUE)

Model their joint patterns and test how results change with different correlations.

6 Mini code sketch

```
mv <- apm_multivariate(es, outcomes = c("yield", "soc", "wue"))
scan <- apm_mvcor_sensitivity(mv, rho = c(0.1, 0.3, 0.5))
apm_multivariate_plot(mv)
```



1 Define outcomes



2 Fit joint model



3 Check p sensitivity



4 Visualize results

7 Take-home message



- ✓ Correlated outcomes deserve joint analysis.
- ✓ Sensitivity checks strengthen conclusions.
- ✓ Multivariate plots make complex results clearer.





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Infographic 18 — Priors and Prior Checks for Bayesian Meta-Analysis

Make assumptions explicit before fitting the Bayesian model



1 Functions emphasized

`f()` `apm_prior()`

`f()` `apm_prior_check()`

2 Why priors matter

- stabilize estimation
- encode weakly informative expectations
- affect extreme or low-information cases
- must be inspected, not hidden



3 Common prior targets



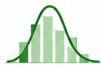
average effect
(pooled effect)



heterogeneity
(between-study SD)



correlations
(among outcomes)



residual scales
(within-study SDs)

4 What prior checks should ask

- are priors too wide or too narrow
- do simulated values look plausible
- are impossible agronomic values being implied
- is domain knowledge respected



5 Applied agronomy example



Crop-yield meta-analysis with weakly informative priors for the pooled effect and heterogeneity.

Pooled effect (average yield difference)
 $\mu \sim \mathcal{N}(\theta, 1^2)$
(weakly informative)

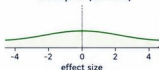
Heterogeneity (between-study SD)
 $\tau \sim \text{Half-Normal}(\theta, 1)$
(weakly informative)

6 Mini code sketch

```
pri <- apm_prior(mv)
chk <- apm_prior_check(mv, prior = pri)
```

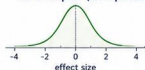
Prior comparison: effect size (e.g., yield difference, $t \text{ ha}^{-1}$)

Broad prior (too wide)



Allows unrealistic extremes.

Narrower prior (more plausible)



Concentrates mass on realistic values.



Choose priors that are weakly informative but rule out impossible values.



Mini prior-checklist

- ☒ Priors reflect plausible agronomic ranges.
- ☒ Simulated values look reasonable.
- ☒ No impossible biophysical values.
- ☒ Sensitivity to reasonable alternatives has been explored.
- ☒ Document choices and rationale.

7 Take-home message



- ✓ priors are assumptions to document
- ✓ prior checks improve transparency
- ✓ plausible priors support more credible inference





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Infographic 19 — Bayesian Fitting, Diagnostics, and Prediction

From posterior estimation to checks of convergence and predictive interpretation



1 Functions emphasized

`f()` `apm_bayes()`

`f()` `apm_bayes_diagnostics()`

`f()` `apm_bayes_predict()`

2 What Bayesian fitting returns



posterior summaries



credible intervals



heterogeneity estimates



predictive distributions

3 Diagnostics to inspect



trace mixing



effective sample size



R-hat near 1



posterior predictive checks

4 Prediction in practice



Predictions describe what may happen in a new agronomic study or setting, incorporating uncertainty from the fitted model.

5 Applied agronomy example



Use irrigation studies to predict the expected yield response (with credible intervals) in a new environment (e.g., different climate, soil, or crop variety).

Bayesian workflow at a glance



Fit

Estimate posteriors with `apm_bayes()`



Diagnose

Check convergence and model fit



Predict

Generate predictive distributions



Interpret

Translate to agronomic insights

6 Mini code sketch

```
fit_b <- apm_bayes(mv, prior = pri)
diag <- apm_bayes_diagnostics(fit_b)
pred <- apm_bayes_predict(fit_b)
```

7 Take-home message



Fitting is only one part of Bayesian workflow.

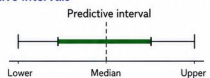
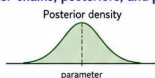
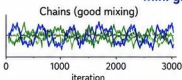


Diagnostics protect against false confidence.



Prediction links inference to future studies.

Mini graphics: chains, posteriors, and predictive intervals





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Infographic 20 — Bayesian Thresholds and Model Comparison

Turn posterior summaries into practical agronomic decisions



1 Functions emphasized

`f()` `apm_bayes_threshold()`

`f()` `apm_bayes_compare()`

`f()` `apm_bayes_predict()`

2 Threshold thinking

- define a practical relevance threshold
- compute posterior support for exceeding it
- report probabilities, not just intervals.



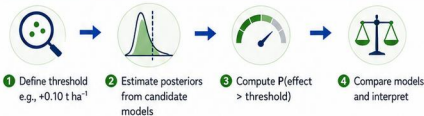
3 Why compare Bayesian models?

- competing prior structures
- competing outcome structures
- competing substantive explanations
- comparison supports transparency.

4 Applied agronomy example



Using a threshold for a practically relevant increase in yield and comparing alternative Bayesian models for fertilizer-response data.



5 What to report



posterior mean or median



credible interval



probability effect > threshold

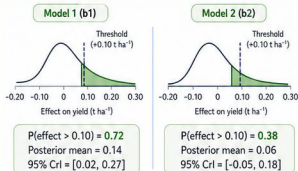


model-comparison summary



practical conclusion

Two-model comparison (example)



Model 1 preferred



Higher probability of meeting the practical threshold.

Bayes factor (b1 vs b2)

3.2

Moderate evidence

6 Mini code sketch

```
thr <- apm_bayes_threshold(fit_b, threshold = 0.10)
cmp <- apm_bayes_compare(fit_b1, fit_b2)
pred <- apm_bayes_predict(fit_b)
```

7 Take-home message



- ✓ thresholds connect statistics to agronomic relevance.
- ✓ Bayesian comparison clarifies uncertainty across models.
- ✓ final decisions should combine probability and practical meaning.





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Infographic 21 — Influence Analysis and Leave-One-Out Diagnostics

Check whether conclusions depend too heavily on a few studies or contrasts



1 Functions emphasized

`f()` `apm_influence()`

`f()` `apm_leave_one_out()`

2 What influence diagnostics ask



Which studies have high leverage?



Which contrasts shift the pooled effect?



Does heterogeneity change after deletion?



Are conclusions driven by a few data points?

3 Key diagnostics to inspect



Cook-type influence



leverage



DFBETAS



change in pooled effect,
 τ^2 , and I^2

4 Leave-one-out workflow



1
Screen
fitted model



2
Compute
influence metrics



3
Refit after
deleting one study



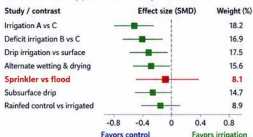
4
Compare pooled
effects and heterogeneity



5
Interpret
stability

5 Applied agronomy example

Irrigation studies on crop yield (t ha^{-1}).
One contrast appears unusually influential.



Most studies are stable, but one contrast meaningfully changes the pooled estimate.

6 Mini code sketch

```
fit <- apm_fit(es)
inf <- apm_influence(fit)
loo <- apm_leave_one_out(fit)
```



7 Take-home message



Influence diagnostics reveal whether evidence is fragile.



Leave-one-out analysis shows how stable conclusions really are.



Sensitivity checks should accompany agronomic interpretation.





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Infographic 22 — GOSH and Publication-Bias Sensitivity

Explore model stability and small-study effects without oversimplifying the cause



1 Functions emphasized

`f()` `apm_gosh()`

`f()` `apm_bias()`

2 What GOSH explores



many subset models



stability of pooled effects



clustering of alternative solutions



detection of unusual model regions

3 What bias methods examine



Egger-type regression



rank correlation



trim-and-fill



selection models



S-values or selection ratios



Copas or limit sensitivity

4 Interpret with caution



Funnel asymmetry is not proof of publication bias.



Small-study effects may also reflect design or context.



Use several sensitivity methods, not a single test.

4 Workflow overview



1 Fit

Fit a primary meta-analysis model.



2 Explore subsets

Use GOSH to examine many subset solutions.



3 Check bias sensitivity

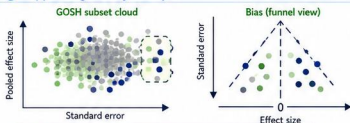
Apply multiple bias sensitivity methods.



4 Interpret carefully

Synthesize evidence with caution.

5 Applied agronomy example



Most subset fits cluster tightly, but a small region suggests instability and small-study concerns.

6 Mini code sketch

```
fit <- apm_fit(es)
g <- apm_gosh(fit)
bias <- apm_bias(fit)
```

7 Take-home message



GOSH helps visualize robustness across many subsets.



Bias sensitivity should combine several complementary methods.



Agronomic interpretation must separate evidence patterns from mechanism claims.





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Infographic 23 — Funnel Contours and Orchard Plots

Communicate uncertainty, subgroup patterns, and small-study visuals more clearly



1 Functions emphasized

`f()` `apm_funnel_contour()`

`f()` `apm_orchard()`

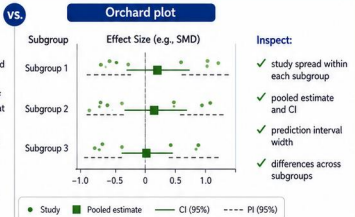
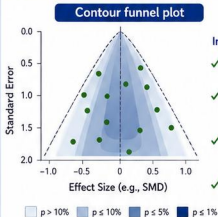
2 What a contour funnel plot shows

- study precision versus effect size
- regions of statistical contour shading
- possible asymmetry patterns
- context for small-study effects

3 What an orchard plot shows

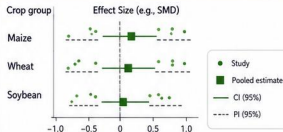
- observed study effects
- pooled subgroup estimates
- confidence interval
- prediction interval
- subgroup comparison at a glance

4 Visual reading guide



5 Applied agronomy example

Effect of a bioinoculant on crop grain yield (standardized mean difference, SMD) across crop groups.



Subgroup means differ, and wide prediction intervals remind us that future results may vary.

6 Mini code sketch

```
fit <- apm_fit(es)
apm_funnel_contour(fit)
apm_orchard(fit, by = crop)
```

7 Take-home message

- ✓ Funnel contours enrich bias visualization.
- ✓ Orchard plots combine effect size, uncertainty, and subgroup structure.
- ✓ Clear visuals improve communication of agronomic meta-analysis.





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Infographic 24 — Exploratory Moderator Screening and Interpretation

Screen many moderators carefully, then translate outputs into audience-ready explanations



1 Functions emphasized

`f()` `apm_moderator_screen()`

`f()` `apm_explain()`

2 What exploratory screening does

- ranks promising moderators
- summarizes variable importance
- supports partial-dependence exploration
- helps prioritize confirmatory modeling

3 Important caution



Screening is exploratory, not confirmatory.

Variable importance is not a causal claim.

Use meta-regression for formal explanation.

4 What explanation output provides

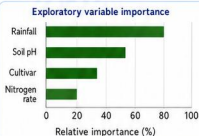
- scientific audience summary
- teaching-oriented interpretation
- extension-friendly communication
- assumptions and cautions made explicit

Workflow overview



5 Applied agronomy example

Explore how rainfall, soil pH, cultivar, and nitrogen rate affect treatment response.



Explanation
(teaching audience)

Rainfall and soil pH show the strongest associations with treatment response. Cultivar has a moderate influence, while nitrogen rate shows a smaller effect in this dataset.

Rainfall and soil pH appear most influential, but confirmation still requires fitted meta-regression.

6 Mini code sketch

```
scr <- apm_moderator_screen(
  es,
  moderators = ~ rainfall +
               soil_pH +
               cultivar +
               dose_n
)

msg <- apm_explain(
  fit,
  audience = "teaching"
)
```

7 Take-home message



- ✓ Screening helps organize complex moderator spaces.
- ✓ Interpretation should be transparent and audience-aware.
- ✓ Exploratory findings must be reported with appropriate caution.





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Infographic 25 — Reporting and Export for Reproducible Meta-Analysis

Turn fitted analyses into reproducible reports, tables, figures, and shareable outputs



1 Functions emphasized

`f()` `apm_report()`

`f()` `apm_export()`

2 What reporting should include

-  data and effect-size provenance
-  model summary and heterogeneity
-  diagnostics and sensitivity analyses
-  tables, figures, and interpretation
-  reproducibility metadata

3 Export formats



html



docx



pdf



csv



xlsx



rds



json



png / svg / tiff

4 Typical closing workflow



1
Fit and
diagnose



2
Build tables
and plots



3
Render
report



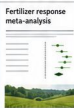
4
Export
reusable files



5
Archive and
share

5 Applied agronomy example

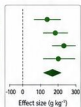
Report page



Summary table

Treatment	ES (g kg ⁻¹)	95% CI
N low	120	[80, 160]
N medium	180	[140, 220]
N high	240	[200, 280]
Heterogeneity	I ² = 52%	

Forest plot



Archive folder



A final agronomic deliverable should combine analysis, visuals, and auditable outputs.

6 Mini code sketch

```
rep <- apm_report(
  fit, "yield_meta.html",
  format = "html")

apm_export(
  apm_table(fit),
  "yield_table.xlsx",
  format = "xlsx")

apm_export(
  apm_forest(fit),
  "forest_plot.tiff",
  format = "tiff")
```

7 Take-home message



Reproducible reporting
turns analysis into a
usable product.



Export tools support
manuscripts, teaching,
and extension.



Clear documentation
closes the meta-analysis
workflow.

